Tree Ensembles for Portfolio Optimization:

Auditing the AI Mirage, Volatility Chains & The First Trade

Random Forests, Gradient Boosting, Permutation Noise Floors & Validated Risk Allocation

Georgios Kompothanasis

FinTech & AI Engineering — Week 6 Sprint
Executive Business School Presentation

Sealed Holdout Evaluation: October 2024 – September 2026

5-Asset Multi-Calendar Universe: SPY (21h UTC, 252d) · EUR/USD (22h UTC, 252d) · BTC (24h UTC, 365d) · ATHEX (15h UTC, 252d) · ETH (24h UTC, 365d)

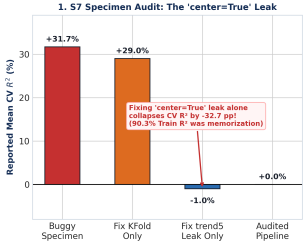
1. Auditing the Junior AI Specimen & Why Single Trees Fail (S7 & S0)

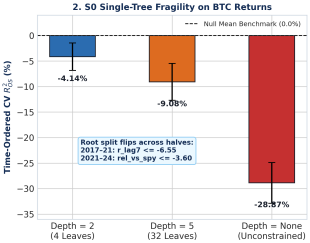
The Junior AI Mirage (agent_draft_tree_portfolio)

- Claimed **+31.7% CV** R^2 and **90.3% Train** R^2, yet delivered **-33.0% return (Sharpe -0.134)** and **99.9% weight in BTC**.
- **Bug 1 (Rule T1 Leak):** `rolling(5, center=True)` peeks 2 days into the future. Fixing it alone collapses CV R^2 from **+31.7% to -1.0%** (-32.7 pp).
- **Bug 4 (Alignment Trap):** Fixing date alignment while keeping in-sample predictions creates a fake **+7.10 Sharpe (+1,930% return)**!

Why Single Trees Are Unstable (S0)

- Splitting BTC training history in half flips the root question completely (`r_lag7` → `rel_vs_spy`).
- Deeper single trees memorize noise: CV R^2_{OS} drops from **-4.14%** ($d = 2$) to **-28.87%** ($d =$ None).





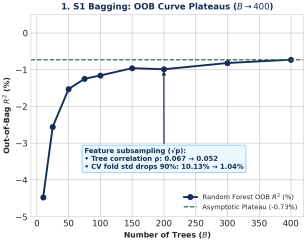
2. Two Ways Ensembles Tame Trees: Bagging vs. Boosting (S1, S2 & S3)

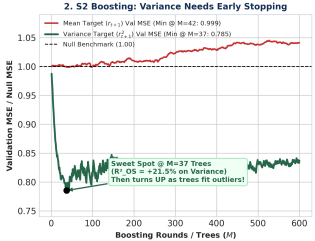
S1 · Random Forest: Disagreement Kills Variance

- Restricting splits to $\sqrt{p}$ features cuts pairwise tree correlation ρ (**0.0666** $\rightarrow$ **0.0523**).
- Cross-fold CV dispersion drops by **90%** (**10.13%** $\rightarrow$ **1.04%**, 0.10×), and OOB R^2 **plateaus** at -0.73%.

S2 · Boosting & S3 · The 2025 Protocol Trap

- **Mean (r_{t+1}):** Boosting validation MSE stays flat at ≈ 1.00 — zero return signal to chase.
- **Variance (r^2_{t+1}):** Validation MSE plunges to **0.785** (**$R^2_{OS} = +21.5\%$) at** $M = 37$, then **bends upward** as later trees fit isolated crash outliers.
- **2025 Protocol Leak:** Shuffled K-Fold + best rung inflates SPY variance CV R^2_{OS} from **+5.39% to +17.16%** (+11.8 pp).





3. Sealed Holdout Showdown & The Risk Agent's Chain (S4)

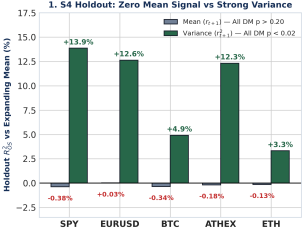
Mean Verdict (Rule T3): Non-Linearity Bought Nothing

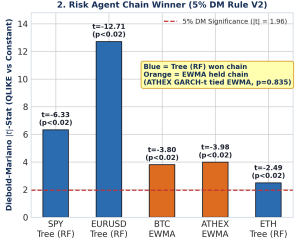
- Across all 5 assets on the 2024–2026 holdout, the best tree ties both the expanding mean and Lasso ($p > 0.20$).
- We pass **zero mean forecasts** to the portfolio (Rule P1).

Risk Agent's Variance Chain (5% DM QLIKE Rule V2)

Walking `Constant` → `EWMA` → `GARCH-t` → `Lasso-HAR` → `Tree`:

- **SPY ($t = -6.33$), EUR/USD ($t = -12.71$), ETH ($t = -2.49$):** Tree (`RF`) beats Constant & simpler models → **Consumes Tree (RF)**.
- **BTC ($t = -3.80$) & ATHEX ($t = -3.98$):** `EWMA` wins; Week 4's `GARCH(1,1)-t` ties `EWMA` on ATHEX ($p = 0.835$) → **Consumes EWMA**.





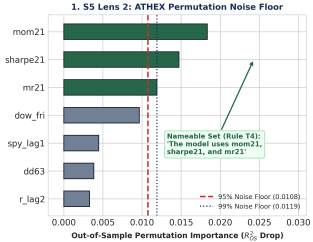
4. Reading the Model: Three Lenses & Permutation Noise Floor (S5)

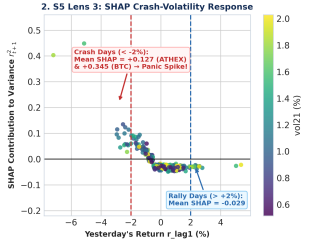
Lens 1 Trap: Impurity Importance Lies on Noise

- When trained on a randomly permuted target (`imp_null`), split-impurity importance still produces a confident bar chart of technical indicators (`macd_hist`, `r_lag2`).

Lens 2 (Noise Floor) & Lens 3 (SHAP Asymmetry)

- **Rule T4 Discipline:** Only features beating the **95% permutation floor (0.0108)** on unseen validation data are named. On ATHEX, **the model uses** `['mom21', 'sharpe21', 'mr21']`.
- **Leverage Effect in SHAP:** Yesterday's return (`r_lag1`) pushes variance strongly **up (+0.127 on ATHEX, +0.345 on BTC)** on crash days ($< -2\%$), but **down (-0.029)** on calm rallies ($> +2\%$).

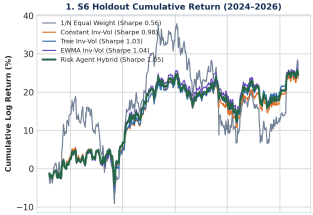


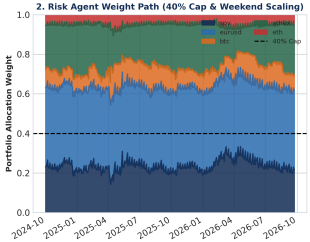


5. The First Trade, Portfolio Scoreboard & Governance Suite (S6 & S8)

Holdout Portfolio Scoreboard (2024-10-01 – 2026-09-23, 40% Cap):

Strategy	Ret%	Vol%	Sharpe	MaxDD%
Equal Weights (1/N)	13.52	24.35	0.555	-31.34
Constant Inv-Vol	11.96	12.18	0.982	-13.34
Tree Inv-Vol	12.65	12.27	1.031	-13.42
EWMA Inv-Vol	12.76	12.28	1.039	-12.17
Risk Agent (Hybrid)	**12.77**	**12.18**	**1.049**	**-12.59**





- **Weekend Scaling:** Scaling 24/7 crypto variance by days = 3 on Mondays generates 4.75%/day baseline turnover even for Constant variance.
- **Stretch & Eval Suite:** SPY regime tree (`rv_week <= 1.80`) cuts vol to **12.03%**; all **40 assertions (8 rules × 5 assets) pass** ✓.